

WASP-43b

R-0441

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_240327 · created 2026-07-19T14:02:10+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460396.6938 +/- 0.0025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.197 +/- 0.058
Transit depth (Rp/Rs)^2	3.88 +/- 2.29 [%]
Orbital Inclination (inc)	84.27 +/- 2.88
Airmass coefficient 1 (a1)	0.423 +/- 0.025
Airmass coefficient 2 (a2)	0.64 +/- 0.46
Scatter in the residuals of the lightcurve fit is	6.22 %
Best Comparison Star	#2 - [385, 37]
Optimal Aperture	6.31
Optimal Annulus	14.19
Transit Duration (day)	0.0571 +/- 0.0079

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.197 ± 0.058 vs 0.1594 (deviation 0.65σ)
Duration fit vs published	0.0571 d vs 0.0512 d (deviation 0.74σ)
Mid-time O–C	-5.3 min
Depth SNR	3.4
Residual scatter	6.22 %

Light curve

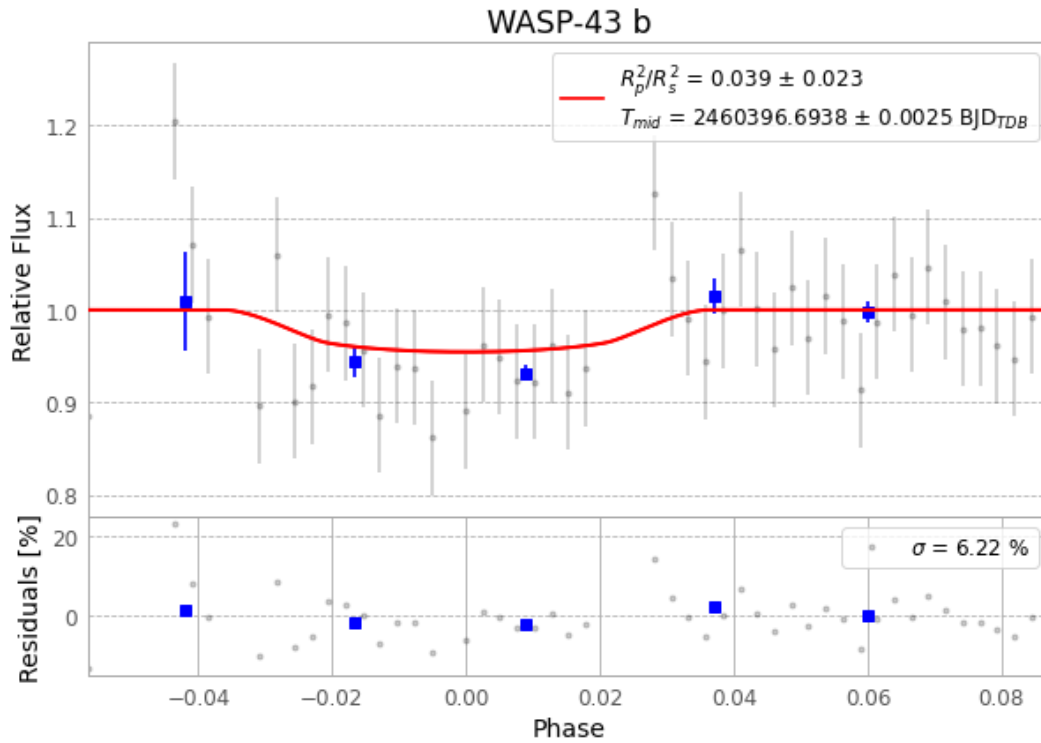


Figure 1 — FinalLightCurve_WASP-43 b_2024-03-26.png (SHA-256 1e9bc3fb1e9bf43a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [463, 231], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 44626d56db034774...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

57 FITS frames (37 MB), WASP-43240327032415.FITS → WASP-43240327061215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-240327.log (0519af1054a8...), FinalLightCurve_WASP-43 b_2024-03-26.png (1e9bc3fb1e9b...), FinalLightCurve_WASP-43 b_2024-03-26.csv (f0cd3a6b9dca...)

Compute resources

Wall time 270.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 14:02Z.